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SENSOR SERIAL NUMBER: 4361
CALIBRATION DATE: 31-Jan-24

SBE 43F OXYGEN CALIBRATION DATA

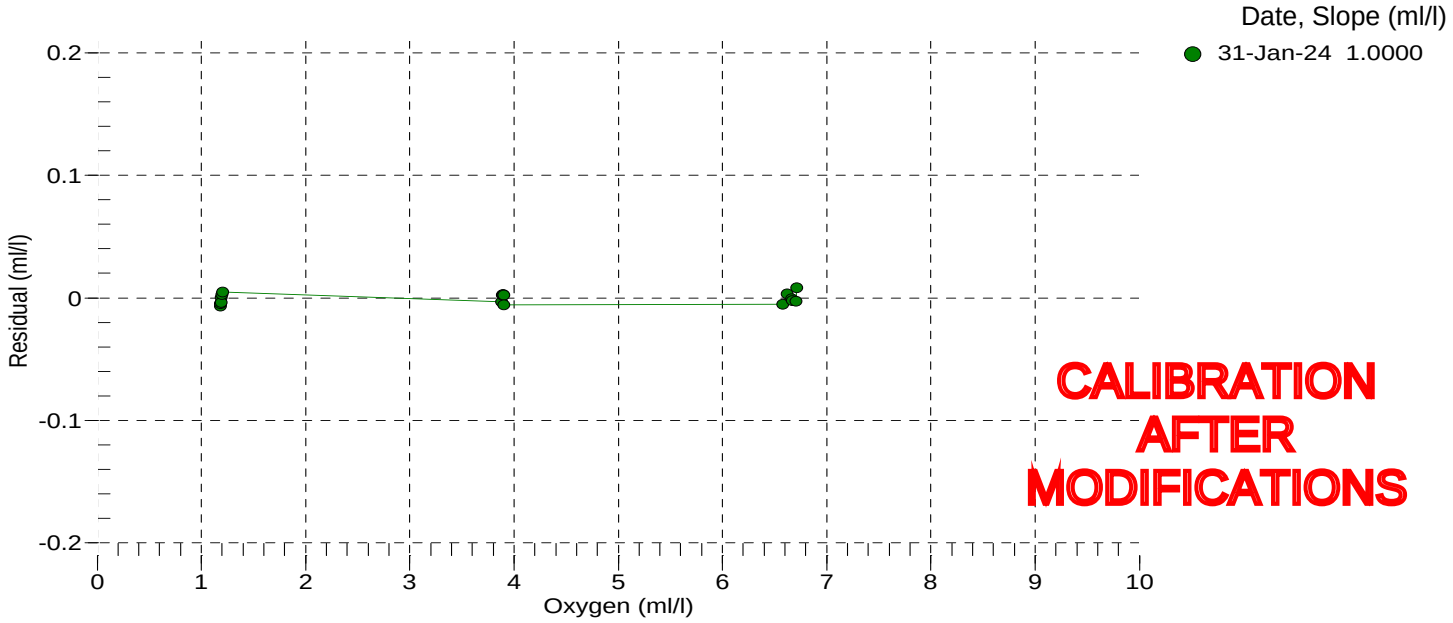
COEFFICIENTS:
Soc = 3.6592e-004
Foffset = -766.69
Tau20 = 1.11
A = -3.6087e-003
B = 1.2683e-004
C = -2.2744e-006
E nominal = 0.036

NOMINAL DYNAMIC COEFFICIENTS
D1 = 1.92634e-4
D2 = -4.64803e-2
H1 = -3.300000e-2
H2 = 5.00000e+3
H3 = 1.45000e+3

BATH OX (ml/l)	BATH TEMP (° C)	BATH SAL (PSU)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT OXYGEN (ml/l)	RESIDUAL (ml/l)
1.18	30.00	0.00	1409.099	1.17	-0.01
1.18	26.00	0.00	1363.017	1.18	-0.00
1.19	12.00	0.00	1209.633	1.19	0.00
1.19	20.00	0.00	1295.988	1.18	-0.00
1.19	6.00	0.00	1148.739	1.20	0.00
1.20	2.00	0.00	1109.437	1.21	0.00
3.88	30.00	0.00	2889.144	3.88	-0.00
3.89	6.00	0.00	2009.111	3.89	0.00
3.89	2.00	0.00	1874.146	3.90	0.00
3.90	26.00	0.00	2738.285	3.90	0.00
3.90	12.00	0.00	2222.831	3.90	0.00
3.90	20.00	0.00	2508.730	3.89	-0.01
6.58	2.00	0.00	2636.044	6.58	-0.01
6.62	6.00	0.00	2881.276	6.62	0.00
6.66	30.00	0.00	4409.357	6.66	-0.00
6.67	12.00	0.00	3255.560	6.67	-0.00
6.71	20.00	0.00	3765.394	6.70	-0.00
6.71	26.00	0.00	4164.468	6.72	0.01

F = instrument output (Hz); T = temperature (°C); S = salinity (PSU); K = temperature (°K)

Oxsol(T,S) = oxygen saturation (ml/l); P = pressure (dbar)

$$\text{Oxygen (ml/l)} = \text{Soc} * (F + \text{Foffset}) * (1.0 + A * T + B * T^2 + C * T^3) * \text{Oxsol}(T,S) * \exp(E * P / K)$$

Residual (ml/l) = instrument oxygen - bath oxygen